Final: sixteen models measured, one packaged, and a target declared infeasible

Results

One run, two architectures re-tuned from scratch on the full 531,431-row corpus, **2,000 search trials**, **sixteen models scored on eight sealed corpora — 128 measured results**, all under one loader, one 58-label catalogue, one fit/calibration carve and one fixed evaluator.

The scoreboard

model	read window	macro F2	micro F1	priority F0.5	doc P / sp / R	p95
pii-cascade-balanced-v2 *(packaged)*	12,000	**0.6614**	0.7255	0.7434	**0.896 / 0.879 / 0.797**	3.92 ms
steady-cascade (arm B)	12,000	0.6497	**0.7313**	**0.7467**	0.889 / 0.883 / 0.753	3.92 ms
fusion-12k (arm C)	12,000	0.4809	0.3647	0.2054	0.616 / 0.001 / 1.000	4.23 ms
fusion-1k (arm A)	1,000	0.4800	0.3652	0.2060	0.616 / 0.001 / 1.000	**1.16 ms**
12 operating points	12,000	0.63–0.66	0.73–0.78	0.75–0.77	unchanged	3.92 ms

The three findings that decided the run

1. The cascade beats the fusion decisively — including on the metric chosen to favour the fusion. macro F2 0.6497 vs 0.4809, on all five corpora that can measure it, with non-overlapping intervals. The fusion's only advantage is recall, and it buys that by firing on 99.99% of documents: **its document precision equals the corpus base rate to four decimals on all three corpora that have negatives**, so its document-level output carries zero information.

2. Source-balancing the gate is a free win. Real-world documents are 7.8% of the fit rows carrying document gold. Equalising their contribution to the loss lifted document recall 0.7532 → 0.7975, macro recall 0.6482 → 0.6938, worst priority-tag recall 0.6524 → 0.7221, and priority gates cleared 25 → 29 of 55 — at no cost in latency and essentially none in precision.

3. The document-level bars are not reachable in this architecture. Best simultaneously achievable is recall **0.623** holding precision ≥ 0.90 and specificity ≥ 0.85 (bar 0.85), or precision **0.766** holding recall ≥ 0.85 (bar 0.90). Three independent lines say this is the representation, not the data:

evidence	result
frontier walk on measured scores	bars sit 0.227 outside it
synthetic near-miss negatives (1,948 verified)	AUC 0.8453 → 0.8461; generated vs real separate at **AUC 1.0000**

evidence	result
real target-distribution data (7,370 labelled)	AUC 0.8832 → 0.8778, flat across a 15× budget range
in-distribution cross-validated ceiling	AUC 0.869 — no better than transferring

Formal verdict recorded: `unlikely`, ceiling ≈ 0.623 against a 0.85 target.

What was packaged

`projects/pii-head-to-head-v1/dist/pii-cascade-balanced-v2/` and its `.zip`, re-scored through its own `tagger.py` after packaging: priority macro F0.5 on `pii_holdout_20k` expected 0.841073, measured 0.841073, **delta 2.5e-08**.

It is packaged as the run's best measured artifact, **not** promoted to `@champion` — it fails hard constraints in both declared policies, and this run's rule is that a gate-failing artifact is not promoted.

TL;DR

- **Sixteen models, 128 sealed results, 2,000 trials.** Full history in `26-08-25_Experiment-Log.xlsx`: 152 Experiment Log rows, 3,384 metric rows with confidence intervals, 5,980 per-tag rows, 8 data-quality rows.
- **The cascade wins**, on every summary metric that prices precision at all, including the recall-weighted one picked to favour its opponent.
- **The fusion is not a detector.** Document precision = base rate; zero information at the document level.
- **Balanced gate adopted**: +0.044 document recall, +0.070 worst priority-tag recall, 4 more gates cleared, free.
- **Target declared infeasible** for this architecture, with the reachable numbers stated. Synthetic data and real data both refuted as remedies.
- **Packaged** `pii-cascade-balanced-v2` with card, zip, checksums and a post-packaging reproduction check.
- **Nothing promoted.** No arm clears its declared gates.

Date	2026-08-26
Author	Ryan Lence
Project	`projects/pii-head-to-head-v1`
Run ID	H1
Scope	2 lineages × 1,000 trials; 16 models × 8 sealed corpora; datagen; feasibility
Request	*"do a head to head on both models. train them on this full dataset … then eval them on each of the datasets in eval … record all the metrics … also the results of each pii tag … each run needs to be recorded in the log"*
Outcome	cascade wins; balanced gate adopted and packaged; document bars infeasible here

Verification discipline

- Cached-feature predictions were proved identical to each model's own `predict()` on sampled real documents — **0 mismatches of 96 per arm**. The first attempt failed arm B with 2 of 96, exposing that `.docx` extraction is read-limit dependent (564 of 4,000 datax documents affected, 14.1%).
- The evaluator was checked against closed-form values before producing any number, including that precision-bearing metrics are `None` on positive-only gold rather than 0.0, and that a prevalence-1.0 corpus emits no document scope.
- The generated corpus was scanned before any document reached training: 52 of 2,000 rejected, including **39 carrying real identifiers under a negative label**.
- The packaged bundle was re-scored through its own entry point: delta 2.5e-08.

Artifacts

Path	What
`dist/pii-cascade-balanced-v2/` + `.zip`	the packaged model, card, checksums
`reports/26-08-25_Experiment-Log.xlsx`	the full history, four tabs
`reports/*.md` · `*.pdf`	four run reports
`reports/26-08-25_head-to-head.commands.txt`	every command, unabridged
`run.json`	128 arms, both decisions, data quality
`evaluations/`	16 model-level arms with all scopes and CIs
`probe/`	feasibility, gate diagnosis, grouped split, augmentation
`tuning/`	all 2,000 trials
`datagen/2000_26082521_claude/`	the generated corpus and its verification
`mlflow.db`	every trial as a tracked run

Model scoreboard

Model	Dataset	F1@5	F1@3	F1@1	Macro F1	Docs/sec	ms/doc
`fusion-1k`	10360_betterdataai_ner_silver_eval_10.36k				0.208	1360.8	0.7
`fusion-1k`	10626_ai4privacy_pii_masking_eval_10.63k				0.424	1360.8	0.7
`fusion-1k`	20000_pii_holdout_20.00k				0.477	1360.8	0.7
`fusion-1k`	30000_pii2_eval_25.15k				0.413	1360.8	0.7
`fusion-1k`	38937_openpii_pii_eval_38.94k				0.510	1360.8	0.7

Model	Dataset	F1@5	F1@3	F1@1	Macro F1	Docs/sec	ms/doc
`fusion-1k`	4000_datax-dualjudge-evalset-1.32k					1360.8	0.7
`fusion-1k`	5617_nemotron_eval_5.36k					1360.8	0.7
`fusion-1k`	6589_govdocs2-dualjudge-eval20-3.53k					1360.8	0.7
`steady-cascade`	10360_betterdataai_ner_silver_eval_10.36k				0.266	314.1	3.2
`steady-cascade`	10626_ai4privacy_pii_masking_eval_10.63k				0.729	314.1	3.2
`steady-cascade`	20000_pii_holdout_20.00k				0.709	314.1	3.2
`steady-cascade`	30000_pii2_eval_25.15k				0.757	314.1	3.2
`steady-cascade`	38937_openpii_pii_eval_38.94k				0.657	314.1	3.2
`steady-cascade`	4000_datax-dualjudge-evalset-1.32k					314.1	3.2
`steady-cascade`	5617_nemotron_eval_5.36k					314.1	3.2
`steady-cascade`	6589_govdocs2-dualjudge-eval20-3.53k					314.1	3.2
`fusion-12k`	10360_betterdataai_ner_silver_eval_10.36k				0.208	270.3	3.7
`fusion-12k`	10626_ai4privacy_pii_masking_eval_10.63k				0.425	270.3	3.7
`fusion-12k`	20000_pii_holdout_20.00k				0.485	270.3	3.7
`fusion-12k`	30000_pii2_eval_25.15k				0.390	270.3	3.7
`fusion-12k`	38937_openpii_pii_eval_38.94k				0.511	270.3	3.7
`fusion-12k`	4000_datax-dualjudge-evalset-1.32k					270.3	3.7
`fusion-12k`	5617_nemotron_eval_5.36k					270.3	3.7
`fusion-12k`	6589_govdocs2-dualjudge-eval20-3.53k					270.3	3.7
`steady-cascade-bal`	10360_betterdataai_ner_silver_eval_10.36k				0.289	314.1	3.2
`steady-cascade-bal`	10626_ai4privacy_pii_masking_eval_10.63k				0.742	314.1	3.2
`steady-cascade-bal`	20000_pii_holdout_20.00k				0.704	314.1	3.2
`steady-cascade-bal`	30000_pii2_eval_25.15k				0.757	314.1	3.2
`steady-cascade-bal`	38937_openpii_pii_eval_38.94k				0.658	314.1	3.2
`steady-cascade-bal`	4000_datax-dualjudge-evalset-1.32k					314.1	3.2
`steady-cascade-bal`	5617_nemotron_eval_5.36k					314.1	3.2
`steady-cascade-bal`	6589_govdocs2-dualjudge-eval20-3.53k					314.1	3.2
`cascade_f65b00`	10360_betterdataai_ner_silver_eval_10.36k				0.247	314.1	3.2
`cascade_f65b00`	10626_ai4privacy_pii_masking_eval_10.63k				0.710	314.1	3.2

Model	Dataset	F1@5	F1@3	F1@1	Macro F1	Docs/sec	ms/doc
`cascade_f65b00`	20000_pii_holdout_20.00k				0.728	314.1	3.2
`cascade_f65b00`	30000_pii2_eval_25.15k				0.765	314.1	3.2
`cascade_f65b00`	38937_openpii_pii_eval_38.94k				0.626	314.1	3.2
`cascade_f65b00`	4000_datax-dualjudge-evalset-1.32k					314.1	3.2
`cascade_f65b00`	5617_nemotron_eval_5.36k					314.1	3.2
`cascade_f65b00`	6589_govdocs2-dualjudge-eval20-3.53k					314.1	3.2
`cascade_f65b10`	10360_betterdataai_ner_silver_eval_10.36k				0.247	314.1	3.2
`cascade_f65b10`	10626_ai4privacy_pii_masking_eval_10.63k				0.711	314.1	3.2
`cascade_f65b10`	20000_pii_holdout_20.00k				0.735	314.1	3.2
`cascade_f65b10`	30000_pii2_eval_25.15k				0.786	314.1	3.2
`cascade_f65b10`	38937_openpii_pii_eval_38.94k				0.634	314.1	3.2
`cascade_f65b10`	4000_datax-dualjudge-evalset-1.32k					314.1	3.2
`cascade_f65b10`	5617_nemotron_eval_5.36k					314.1	3.2
`cascade_f65b10`	6589_govdocs2-dualjudge-eval20-3.53k					314.1	3.2
`cascade_f65b25`	10360_betterdataai_ner_silver_eval_10.36k				0.247	314.1	3.2
`cascade_f65b25`	10626_ai4privacy_pii_masking_eval_10.63k				0.722	314.1	3.2
`cascade_f65b25`	20000_pii_holdout_20.00k				0.743	314.1	3.2
`cascade_f65b25`	30000_pii2_eval_25.15k				0.786	314.1	3.2
`cascade_f65b25`	38937_openpii_pii_eval_38.94k				0.647	314.1	3.2
`cascade_f65b25`	4000_datax-dualjudge-evalset-1.32k					314.1	3.2
`cascade_f65b25`	5617_nemotron_eval_5.36k					314.1	3.2
`cascade_f65b25`	6589_govdocs2-dualjudge-eval20-3.53k					314.1	3.2
`cascade_f65b50`	10360_betterdataai_ner_silver_eval_10.36k				0.282	314.1	3.2
`cascade_f65b50`	10626_ai4privacy_pii_masking_eval_10.63k				0.742	314.1	3.2
`cascade_f65b50`	20000_pii_holdout_20.00k				0.754	314.1	3.2
`cascade_f65b50`	30000_pii2_eval_25.15k				0.786	314.1	3.2
`cascade_f65b50`	38937_openpii_pii_eval_38.94k				0.663	314.1	3.2
`cascade_f65b50`	4000_datax-dualjudge-evalset-1.32k					314.1	3.2
`cascade_f65b50`	5617_nemotron_eval_5.36k					314.1	3.2

Model	Dataset	F1@5	F1@3	F1@1	Macro F1	Docs/sec	ms/doc
`cascade_f65b50`	6589_govdocs2-dualjudge-eval20-3.53k					314.1	3.2
`cascade_f70b00`	10360_betterdataai_ner_silver_eval_10.36k				0.274	314.1	3.2
`cascade_f70b00`	10626_ai4privacy_pii_masking_eval_10.63k				0.707	314.1	3.2
`cascade_f70b00`	20000_pii_holdout_20.00k				0.713	314.1	3.2
`cascade_f70b00`	30000_pii2_eval_25.15k				0.761	314.1	3.2
`cascade_f70b00`	38937_openpii_pii_eval_38.94k				0.638	314.1	3.2
`cascade_f70b00`	4000_datax-dualjudge-evalset-1.32k					314.1	3.2
`cascade_f70b00`	5617_nemotron_eval_5.36k					314.1	3.2
`cascade_f70b00`	6589_govdocs2-dualjudge-eval20-3.53k					314.1	3.2
`cascade_f70b10`	10360_betterdataai_ner_silver_eval_10.36k				0.274	314.1	3.2
`cascade_f70b10`	10626_ai4privacy_pii_masking_eval_10.63k				0.710	314.1	3.2
`cascade_f70b10`	20000_pii_holdout_20.00k				0.720	314.1	3.2
`cascade_f70b10`	30000_pii2_eval_25.15k				0.776	314.1	3.2
`cascade_f70b10`	38937_openpii_pii_eval_38.94k				0.641	314.1	3.2
`cascade_f70b10`	4000_datax-dualjudge-evalset-1.32k					314.1	3.2
`cascade_f70b10`	5617_nemotron_eval_5.36k					314.1	3.2
`cascade_f70b10`	6589_govdocs2-dualjudge-eval20-3.53k					314.1	3.2
`cascade_f70b25`	10360_betterdataai_ner_silver_eval_10.36k				0.274	314.1	3.2
`cascade_f70b25`	10626_ai4privacy_pii_masking_eval_10.63k				0.714	314.1	3.2
`cascade_f70b25`	20000_pii_holdout_20.00k				0.724	314.1	3.2
`cascade_f70b25`	30000_pii2_eval_25.15k				0.776	314.1	3.2
`cascade_f70b25`	38937_openpii_pii_eval_38.94k				0.647	314.1	3.2
`cascade_f70b25`	4000_datax-dualjudge-evalset-1.32k					314.1	3.2
`cascade_f70b25`	5617_nemotron_eval_5.36k					314.1	3.2
`cascade_f70b25`	6589_govdocs2-dualjudge-eval20-3.53k					314.1	3.2
`cascade_f70b50`	10360_betterdataai_ner_silver_eval_10.36k				0.273	314.1	3.2
`cascade_f70b50`	10626_ai4privacy_pii_masking_eval_10.63k				0.732	314.1	3.2
`cascade_f70b50`	20000_pii_holdout_20.00k				0.733	314.1	3.2
`cascade_f70b50`	30000_pii2_eval_25.15k				0.776	314.1	3.2

Model	Dataset	F1@5	F1@3	F1@1	Macro F1	Docs/sec	ms/doc
`cascade_f70b50`	38937_openpii_pii_eval_38.94k				0.660	314.1	3.2
`cascade_f70b50`	4000_datax-dualjudge-evalset-1.32k					314.1	3.2
`cascade_f70b50`	5617_nemotron_eval_5.36k					314.1	3.2
`cascade_f70b50`	6589_govdocs2-dualjudge-eval20-3.53k					314.1	3.2
`cascade_f75b00`	10360_betterdataai_ner_silver_eval_10.36k				0.266	314.1	3.2
`cascade_f75b00`	10626_ai4privacy_pii_masking_eval_10.63k				0.716	314.1	3.2
`cascade_f75b00`	20000_pii_holdout_20.00k				0.696	314.1	3.2
`cascade_f75b00`	30000_pii2_eval_25.15k				0.748	314.1	3.2
`cascade_f75b00`	38937_openpii_pii_eval_38.94k				0.645	314.1	3.2
`cascade_f75b00`	4000_datax-dualjudge-evalset-1.32k					314.1	3.2
`cascade_f75b00`	5617_nemotron_eval_5.36k					314.1	3.2
`cascade_f75b00`	6589_govdocs2-dualjudge-eval20-3.53k					314.1	3.2
`cascade_f75b10`	10360_betterdataai_ner_silver_eval_10.36k				0.266	314.1	3.2
`cascade_f75b10`	10626_ai4privacy_pii_masking_eval_10.63k				0.716	314.1	3.2
`cascade_f75b10`	20000_pii_holdout_20.00k				0.700	314.1	3.2
`cascade_f75b10`	30000_pii2_eval_25.15k				0.758	314.1	3.2
`cascade_f75b10`	38937_openpii_pii_eval_38.94k				0.647	314.1	3.2
`cascade_f75b10`	4000_datax-dualjudge-evalset-1.32k					314.1	3.2
`cascade_f75b10`	5617_nemotron_eval_5.36k					314.1	3.2
`cascade_f75b10`	6589_govdocs2-dualjudge-eval20-3.53k					314.1	3.2
`cascade_f75b25`	10360_betterdataai_ner_silver_eval_10.36k				0.266	314.1	3.2
`cascade_f75b25`	10626_ai4privacy_pii_masking_eval_10.63k				0.721	314.1	3.2
`cascade_f75b25`	20000_pii_holdout_20.00k				0.702	314.1	3.2
`cascade_f75b25`	30000_pii2_eval_25.15k				0.758	314.1	3.2
`cascade_f75b25`	38937_openpii_pii_eval_38.94k				0.650	314.1	3.2
`cascade_f75b25`	4000_datax-dualjudge-evalset-1.32k					314.1	3.2
`cascade_f75b25`	5617_nemotron_eval_5.36k					314.1	3.2
`cascade_f75b25`	6589_govdocs2-dualjudge-eval20-3.53k					314.1	3.2
`cascade_f75b50`	10360_betterdataai_ner_silver_eval_10.36k				0.266	314.1	3.2

Model	Dataset	F1@5	F1@3	F1@1	Macro F1	Docs/sec	ms/doc
`cascade_f75b50`	10626_ai4privacy_pii_masking_eval_10.63k				0.729	314.1	3.2
`cascade_f75b50`	20000_pii_holdout_20.00k				0.709	314.1	3.2
`cascade_f75b50`	30000_pii2_eval_25.15k				0.757	314.1	3.2
`cascade_f75b50`	38937_openpii_pii_eval_38.94k				0.657	314.1	3.2
`cascade_f75b50`	4000_datax-dualjudge-evalset-1.32k					314.1	3.2
`cascade_f75b50`	5617_nemotron_eval_5.36k					314.1	3.2
`cascade_f75b50`	6589_govdocs2-dualjudge-eval20-3.53k					314.1	3.2

Measured 2026-08-25 · n=10,360, n=10,626, n=20,000, n=30,000, n=38,937, n=4,000, n=5,617, n=6,589.